Reinforcement Learning

Our formulation of the problem based on [3] is to find $\pi : \mathcal{S} \to \mathcal{P}(\mathcal{A})$ ($\mathcal{P}$ is a p-measure) such that

$$\max_{\pi} J(\pi) = \max_{\pi} \mathsf{E}_{\tau \sim p(\tau|\pi)} \left[\sum_{i=0}^{\infty} \gamma^i \underbrace{R(\boldsymbol{s}_i, a_i, \boldsymbol{s}_{i+1})}_{r_i} \right]$$

$$p(\tau \mid \pi) = p(\boldsymbol{s}_0) \cdot \prod_{t=0}^{\infty} \pi(a_t|\boldsymbol{s}_t) \cdot P_{a_t}(\boldsymbol{s}_{t+1}, \boldsymbol{s}_t)$$

$$\boldsymbol{s}_{t+1} \sim P_{a_t}(\cdot, \boldsymbol{s}_t) \quad a_t \sim \pi(\cdot|\boldsymbol{s}_t)$$

Where τ is a trajectory of the system, which consists of the states, actions and rewards at each time step.

$$\boldsymbol{\tau}_t = \left\{ \boldsymbol{s}_{t+i}, a_{t+i}, r_{t+i}, \boldsymbol{s}_{t+i+1} \right\}_{i=0}^{\infty} \Rightarrow J(\pi) = \mathsf{E}\left[R(\boldsymbol{\tau}_0) \right]$$

Policy Gradient Methods

Let π_θ be a parameterized policy with $\theta \in \mathbb{R}^d$.

$$\begin{aligned} \nabla_\theta J(\theta) &= \nabla_\theta \mathsf{E}_{\tau \sim p(\tau|\theta)} \left[R(\boldsymbol{\tau}) \right] \\ &= \int_{\mathcal{T}} R(\boldsymbol{\tau}) \cdot \underbrace{\nabla_\theta \log(p(\boldsymbol{\tau} \mid \theta)) \cdot p(\boldsymbol{\tau} \mid \theta)}_{\nabla_\theta p(\boldsymbol{\tau}|\theta)} \, \mathrm{d}\boldsymbol{\tau} \\ &= \mathsf{E}_{\tau \sim p(\tau|\theta)} \left[R(\boldsymbol{\tau}) \cdot \nabla_\theta \log(p(\boldsymbol{\tau} \mid \theta)) \right] \end{aligned}$$

The Logarithm applied to the distribution $p(\boldsymbol{\tau} \mid \theta)$:

$$\begin{aligned} &\nabla_\theta \log \left[p(\boldsymbol{s}_0) \cdot \prod_{k=0}^{T} \pi_\theta(a_k|\boldsymbol{s}_k) \cdot P_{a_k}(\boldsymbol{s}_{k+1}, \boldsymbol{s}_k) \right] \\ &= \sum_{k=0}^{T} \nabla_\theta \log \left[\pi_\theta(a_k|\boldsymbol{s}_k) \right] \end{aligned}$$

Let $R(\boldsymbol{\tau}) = \sum_{i=0}^{T-1} r_i + r_T$ where $r_i = R(\boldsymbol{s}_i, a_i, \boldsymbol{s}_{i+1})$

$$\begin{aligned} &\mathsf{E}_{\tau \sim p(\tau|\theta)} \left[R(\boldsymbol{\tau}) \cdot \sum_{k=0}^{T} \nabla_\theta \log \left[\pi_\theta(a_k|\boldsymbol{s}_k) \right] \right] \\ &= \sum_{i=0}^{T} \sum_{k=0}^{T} \mathsf{E}_\tau \left[r_i \cdot \nabla_\theta \log \left[\pi_\theta(a_k|\boldsymbol{s}_k) \right] \right] \end{aligned}$$

For fixed $i < k$ the law of total expectation yields:

$$\begin{aligned} &\mathsf{E}_{\tau \sim p(\tau|\theta)} \left[r_i \cdot \nabla_\theta \log \left[\pi_\theta(a_k|\boldsymbol{s}_k) \right] \right] \\ &= \mathsf{E}_{a_k, \boldsymbol{s}_k} \left[\nabla_\theta \log \left[\pi_\theta(a_k|\boldsymbol{s}_k) \right] \cdot \underbrace{\mathsf{E}_\tau \left[r_i \mid a_k, \boldsymbol{s}_k \right]}_{\mathsf{E}_\tau[r_i|\boldsymbol{s}_k] = g(\boldsymbol{s}_k)} \right] \\ &= \mathsf{E}_{\boldsymbol{s}_k} \left[\mathsf{E}_{a_k|\boldsymbol{s}_k} \left[\left[g(\boldsymbol{s}_k) \cdot \nabla_\theta \log \left(\pi_\theta(a_k|\boldsymbol{s}_k) \right) \right] \mid \boldsymbol{s}_k \right] \right] \end{aligned}$$

Now using $g(\boldsymbol{s}_k)$ is independent of a_k and therefore

$$\begin{aligned} &\mathsf{E}_{\boldsymbol{s}_k} \left[g(\boldsymbol{s}_k) \mathsf{E}_{a_k|\boldsymbol{s}_k} \left[\left[\nabla_\theta \log \left(\pi_\theta(a_k|\boldsymbol{s}_k) \right) \right] \mid \boldsymbol{s}_k \right] \right] \\ &= \mathsf{E}_{\boldsymbol{s}_k} \left[g(\boldsymbol{s}_k) \cdot \int_{\mathcal{A}} \underbrace{\nabla_\theta \log \left(\pi_\theta(a|\boldsymbol{s}_k) \right) \cdot \pi_\theta(a|\boldsymbol{s}_k)}_{\nabla_\theta \pi_\theta(a|\boldsymbol{s}_k)} \, \mathrm{d}a \right] \\ &= \nabla_\theta \underbrace{\int_{\mathcal{A}} \pi_\theta(a|\boldsymbol{s}_k) \mathrm{d}a}_{=1} = 0 \end{aligned}$$

We conclude that for $i < k$ the expectation is zero and we denote $R_t = \sum_{k=t}^{T} r_k$ and obtain the formula:

$$\nabla_\theta J(\theta) = \sum_{t=0}^{T} \mathsf{E}_{\tau \sim p(\tau|\theta)} \left[\nabla_\theta \log \left(\pi_\theta(a_t|\boldsymbol{s}_t) \right) \cdot R_t \right]$$

Conditioned on $(\boldsymbol{s}_t, a_t)$ we get for $\nabla_\theta J(\theta)$:

$$\begin{aligned} \nabla_\theta J(\theta) &= \mathsf{E}\left[\mathsf{E}\left[\nabla_\theta \log \left(\pi_\theta(a_t|\boldsymbol{s}_t) \right) \cdot R_t \mid a_t, \boldsymbol{s}_t \right] \right] \\ &= \sum_{t=0}^{T} \mathsf{E}_{a_t, \boldsymbol{s}_t} \left[\nabla_\theta \log \left(\pi_\theta(a_t|\boldsymbol{s}_t) \right) \underbrace{\mathsf{E}_{\tau \sim p(\tau|\theta)} \left[R_t \mid a_t, \boldsymbol{s}_t \right]}_{Q^{\pi_\theta}(\boldsymbol{s}_t, a_t)} \right] \end{aligned}$$

Application on Control Problems

We parametrize the control $u(x,t) = u(t, x \mid \theta)$ and relax the problem by considering $a_t \sim \pi(u(\boldsymbol{s}_t, \theta))$.

Control Problem	RL Problem
$x(t)$	$\boldsymbol{s}_t = \begin{bmatrix} t \\ x(t) \end{bmatrix}$
$u(t)$	$a_t \sim \mathcal{N}(u(\boldsymbol{s}_t, \theta), \sigma)$
$\int_0^T L(x,u) \, \mathrm{d}t + \Phi(x(T))$	$\mathsf{E}_\tau \left[\sum_{i=0}^{N} r_i \right]$
$x'(t) = f(x,u)$	$\boldsymbol{s}_{t+1} \sim P_{a_t}(\cdot, \boldsymbol{s}_t)$

1. Let $\mathcal{U} = \{u : [0,T] \to \mathbb{R} \mid u \text{ measurable}\}$

$$\begin{aligned} &\min_{u \in \mathcal{U}} \int_0^T L(X_t, u) \, \mathrm{d}t + \Phi(X_T), \\ &\mathrm{d}X_t = f(X_t, u(t)) \, \mathrm{d}t + \sigma \, \mathrm{d}W_t, \quad X_0 = x_0 \end{aligned}$$

2. Relaxation of the Problem: $\mathcal{U} \subset \mathcal{U}_{stoch}$

$$\begin{aligned} \mathcal{U}_{stoch} &= \{\pi : \mathcal{S} \to \mathcal{P}(\mathbb{R}) \mid \pi(\cdot|\boldsymbol{s}_t) \text{ is p-measure}\} \\ u &: \mathbb{R}^{n_x} \times [0,T] \to \mathbb{R} \Rightarrow \pi(\cdot|\boldsymbol{s}_t) = \delta_{u(\boldsymbol{s}_t)} \end{aligned}$$

3. Discretize: $\boldsymbol{a} = \{a_i\}_{i=0}^{N-1}$ $a_i \sim \pi(u(\boldsymbol{s}_i, \theta))$

$$\log \left[\underbrace{\prod_{i=0}^{N-1} \pi_\theta(a_i|\boldsymbol{s}_i)}_{\pi_\theta(\boldsymbol{a})} \right] = \sum_{i=0}^{N-1} \log \left[\pi_\theta(a_i|\boldsymbol{s}_i) \right]$$

4. Differentiation of the objective function:

$$\nabla_\theta \mathsf{E}\left[\int_0^T L(X_t, a_t)\,\mathrm{d}t + \Phi(X_T)\right]$$
$$= \int_{\mathcal{A}} \underbrace{\pi_\theta(a)\cdot\nabla_\theta \log(\pi_\theta(a))}_{\nabla_\theta \pi_\theta(a)} \left[\Delta t \cdot \sum_{i=0}^{N-1} r_i + \Phi(X_T)\right] \mathrm{d}a$$
$$= \mathsf{E}\left[\sum_{i=0}^{N-1} \nabla_\theta \log(\pi_\theta(a_i)) \cdot C(a_0, \ldots, a_{N-1})\right]$$

5. Policy Gradient Theorem: $C(a_i, \ldots, a_{N-1}) := R_i$

$$\nabla_\theta \mathsf{E}\left[C(\boldsymbol{a})\right] = \mathsf{E}\left[\sum_{i=0}^{N-1} \nabla_\theta \log\left(\pi_\theta(a_i)\right) \cdot R_i\right]$$
$$\approx \frac{1}{M}\sum_{m=1}^{M}\left[\sum_{i=0}^{N-1} \nabla_\theta \log\left(\pi_\theta(a_i^{(m)})\right) \cdot R_i^{(m)}\right]$$

5. Gradient method [1] with $M = 1$:

$$\theta_{k+1} = \theta_k - \alpha_k \nabla_\theta J(\theta_k)$$
$$X_{t_{i+1}} = X_{t_i} + f(X_{t_i}, u_i)\Delta t + \sigma Z_i$$

Numerical Example

We consider the following example with $X_0 = 0$:

$$\min_u \int_0^1 X_t + u_t^2\,\mathrm{d}t + X_1^2 \quad \mathrm{d}X_t = X_t + u_t + 1,$$

and use the stochastic Relaxation:

$$\boldsymbol{s}_t = \begin{bmatrix} t \\ X_t \end{bmatrix} \quad a_t \sim \mathcal{N}\left(u(\boldsymbol{s}_t, \theta), b\right)$$

with the parametrization:

$$u(\boldsymbol{s}_t, \theta) = W_N \cdot \tanh\left(W_1 \cdot \begin{bmatrix} t \\ X_t \end{bmatrix} + b_1\right) + b_N$$

with $W_1 \in \mathbb{R}^{n\times 2}, \boldsymbol{b}_1 \in \mathbb{R}^n, W_N \in \mathbb{R}^{1\times n}, b_N \in \mathbb{R}$. Where $\theta \in \mathbb{R}^{4n+1}$ and n is a Hyperparameter.
For the Reward we compute:

$$r_i = \left(X_{t_i} + a_{t_i}^2\right)\Delta t \quad r_N = X_{t_N}^2 \Rightarrow R_k := \sum_{i=k}^{N} r_i$$

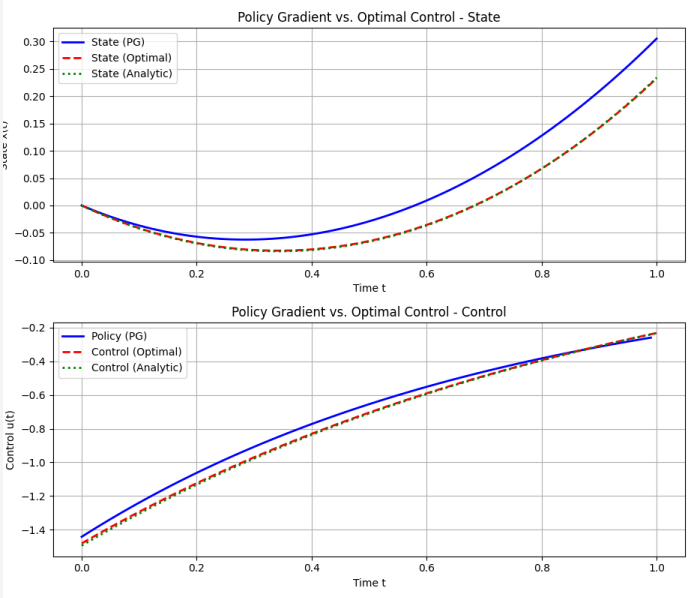
For $M = 1$ the estimator of our Gradient is given by

$$\nabla_\theta J(\theta) \approx \hat{g} = \sum_{i=0}^{N-1} \nabla_\theta \log(\pi_\theta(a_{t_i} \mid \boldsymbol{s}_{t_i})) \cdot R_i$$
$$= \sum_{i=0}^{N-1} \frac{a_{t_i} - u(\boldsymbol{s}_{t_i}, \theta)}{b} \cdot R_i$$

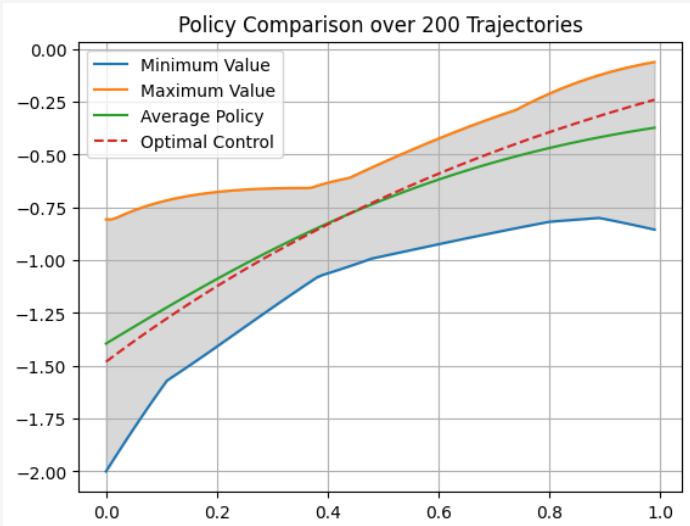
Using the dynamic for Simulation:

$$X_{t_{i+1}} = X_{t_i} + \Delta t\left(X_{t_i} + a_{t_i} + 1\right) \quad X_{t_0} = 0$$

Application of the standard REINFORCE Algorithm with Monte Carlo Sampling [1] leads to



The following results show the high variance in the standard Reinforce-Algorithm as described in [2].



References

[1] Aske Plaat. *Deep Reinforcement Learning.* Springer, 2022. ISBN: 978-981-19-0637-4.

[2] Benjamin Recht. "A Tour of Reinforcement Learning: The View from Continuous Control". In: (2018).

[3] Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction.* MIT Press, 2018.